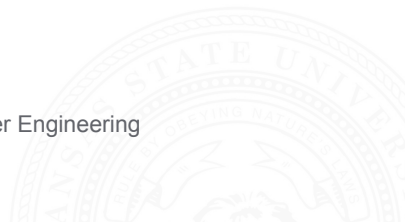


Generation Mix Planning II

From a capacity target to an economically screened generation mix

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One capacity target, many possible mixes

L03 determined how much installed capacity is required

Capacity requirement from L03

$$4380(1 + 0.15) = \boxed{5037 \text{ MW}}$$

Peak demand plus the course planning reserve margin.

A motivating comparison

- A gas turbine is less expensive to build but more expensive to operate.
- A combined-cycle unit is more expensive to build but uses less fuel per MWh.

Which one costs less over a year?

The capacity target specifies how much to build. Operating duration determines which technology has the lower annual cost.

Planning questions

1. Economic comparison

What does the mix cost over a year?

- construction and fixed O&M
- fuel and variable O&M
- different operating durations

2. Capacity mapping

How do cost choices become a mix?

- start with the L03 load-duration curve
- account for wind and solar capacity
- map cost crossovers to MW blocks

PART I

Screening Curves

How annual fixed and variable costs change with operating duration

Why build a screening curve?

1. Identify unit annual cost

$$c_i(T) = c_i^{\text{fix}} + c_i^{\text{var}}T$$

Unit: $c_i(T)$ is in \$/MW-yr.

T : annual operating hours.

$c_i(T)$: annual cost per installed MW.

One consistent MW/MWh basis is used throughout.

$$c_i^{\text{fix}} \text{ [$/MW}\cdot\text{yr]}, \quad c_i^{\text{var}} \text{ [$/MWh]}.$$

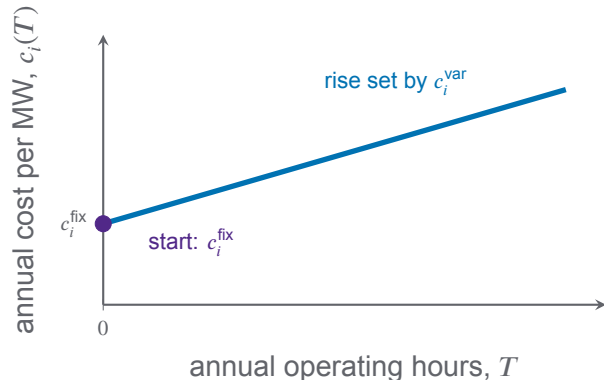
2. Scale to total annual cost

$$C_i^{\text{ann}} = P_i^{\text{cap}} c_i(T_i) \quad [$/\text{yr}]$$

- P_i^{cap} : installed capacity in MW
- T_i : expected annual operating hours
- Read $c_i(T_i)$, then multiply by P_i^{cap} .

Operating duration determines annual cost per MW. Installed capacity scales this value to total annual cost.

One technology, one screening curve



Heads up

- Each candidate technology has one line.
- Fixed cost determines where the line starts.
- Variable cost determines how fast it rises.

The next slides calculate each curve's starting point and slope, followed by a two-curve comparison.

Concept 1: annual fixed-cost coefficient

Annualize capital cost and fixed O&M

$$c_i^{\text{fix}} = \alpha I_i + \text{FOM}_i \quad [\$/\text{MW} \cdot \text{yr}]$$

Inputs

- I_i : capital cost in \$/MW
- α : annual carrying-charge rate.
The project specifies 0.12 yr^{-1}
- FOM_i : fixed O&M in \$/MW-yr

Result

- FOM_i is one cost input.
- c_i^{fix} combines annualized capital and fixed O&M.
- $c_i(0) = c_i^{\text{fix}}$, so it is the intercept.

c_i^{fix} is the screening curve's fixed-cost coefficient and vertical intercept.

Example 1: annual fixed cost

1 Simple-cycle gas turbine (GT)

$$\begin{aligned}c_{GT}^{\text{fix}} &= 0.12(1,700,000) + 13,000 \\ &= 217,000 \text{ \$/MW}\cdot\text{yr.}\end{aligned}$$

2 Combined-cycle gas turbine (CC)

$$\begin{aligned}c_{CC}^{\text{fix}} &= 0.12(2,200,000) + 20,000 \\ &= 284,000 \text{ \$/MW}\cdot\text{yr.}\end{aligned}$$

The simple-cycle gas turbine begins with the lower annual fixed-cost coefficient:
 $217,000 < 284,000$.

Concept 2: variable operating cost

Combine fuel use and variable O&M on the MWh basis

$$c_i^{\text{var}} = \text{VOM}_i + \text{HR}_i p_i^{\text{fuel}} \quad [$/\text{MWh}]$$

- VOM_i : variable O&M in $$/\text{MWh}$
- HR_i : heat rate in MMBtu/MWh
- p_i^{fuel} : fuel price in $$/\text{MMBtu}$

Graph interpretation

- c_i^{var} is the screening-curve slope.
- A lower heat rate reduces the fuel-cost contribution.

c_i^{var} gives cost per MWh, and annual variable cost grows with electricity production.

Example 2: variable operating cost

1 Simple-cycle gas turbine (GT)

$$\begin{aligned}c_{GT}^{\text{var}} &= 9.56 + (10.447)(3.00) \\ &= 40.901 \text{ \$/MWh.}\end{aligned}$$

2 Combined-cycle gas turbine (CC)

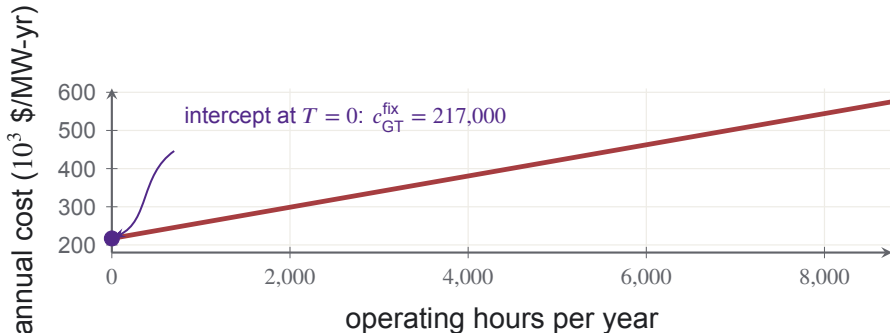
$$\begin{aligned}c_{CC}^{\text{var}} &= 5.05 + (7.000)(3.00) \\ &= 26.05 \text{ \$/MWh.}\end{aligned}$$

The combined-cycle unit has the lower variable operating-cost coefficient: $26.05 < 40.901$.

Concept 3: build one screening curve

Simple-cycle gas turbine: combine its fixed and variable coefficients

$$c_{GT}(T) = 217,000 + 40.901T \quad [\$/\text{MW}\cdot\text{yr}]$$



For one screening curve, c_{GT}^{fix} sets the starting point and c_{GT}^{var} sets the slope.

Concept 4: curve crossover

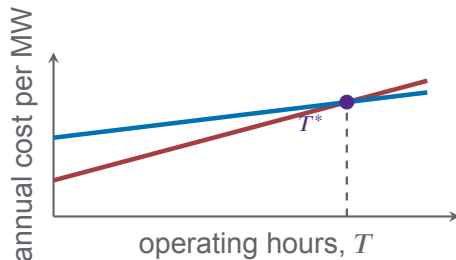
A crossover compares two complete screening curves

Equality at one duration

$$c_i(T^*) = c_j(T^*)$$

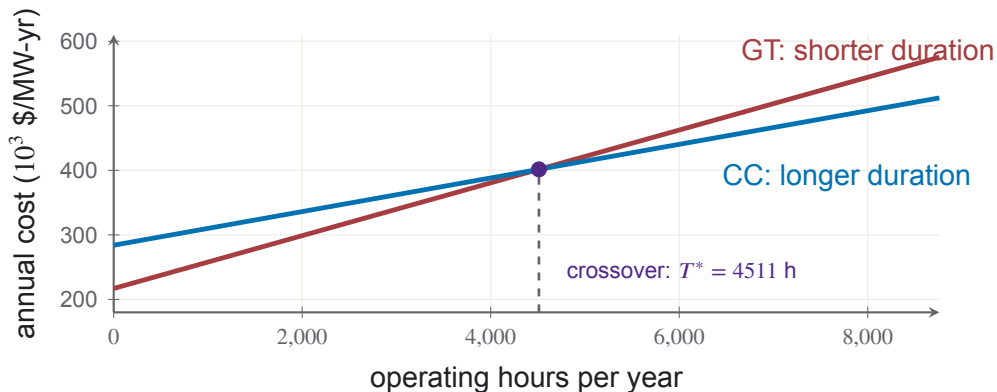
$$T^* = \frac{c_j^{\text{fix}} - c_i^{\text{fix}}}{c_i^{\text{var}} - c_j^{\text{var}}}$$

A relevant crossover must fall between 0 and 8760 h/yr.



A crossover occurs where two complete screening curves have the same annual cost.

Example 3: simple cycle vs. combined cycle



The curves cross at $T^* = 4511$ h/yr. GT has the lower annual cost at shorter durations, and CC has the lower annual cost at longer durations.

Four technologies for the case

Part II uses four resources from the case

Technology	Abbr.	What is given or compared	Role in Part II
Onshore wind	W	500 MW teaching-case block	renewable contribution
Solar PV	PV	300 MW teaching-case block	renewable contribution
Combined-cycle gas turbine	CC	$c_{CC}(T) = 284,000 + 26.05T$	longer-duration load
Simple-cycle gas turbine	GT	$c_{GT}(T) = 217,000 + 40.901T$	shorter-duration load

Wind and solar enter as given capacity blocks. The GT–CC crossover identifies continuous dispatchable capacity requirements for unit rounding.

PART II

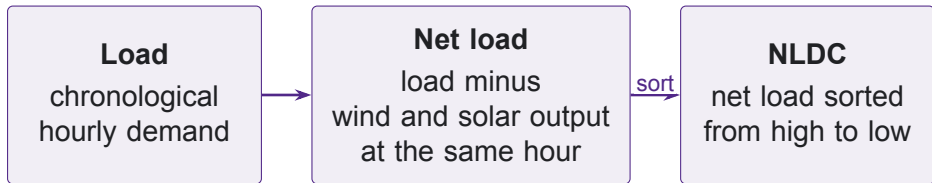
Simplified Generation-Mix Planning

From the load-duration curve to a candidate generation mix

(Simplified teaching workflow. Industry planning includes additional studies.)

From load to net load

Define the curves before applying the teaching case



LDC: chronological load sorted from high to low.

Generic planning workflow

1. Form the NLDC

Calculate hourly net load, then sort it.

2. Map and size

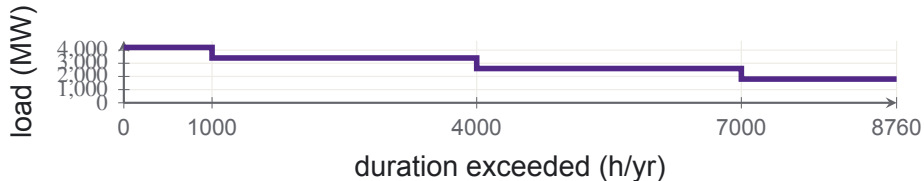
Map the crossover to MW.
Round units to cover the load.

3. Close the PRM gap

After load coverage, compare credited capacity with the PRM target.

Case inputs

First objective: cover the 4200 MW load curve with the four technologies



Renewable capacity plan

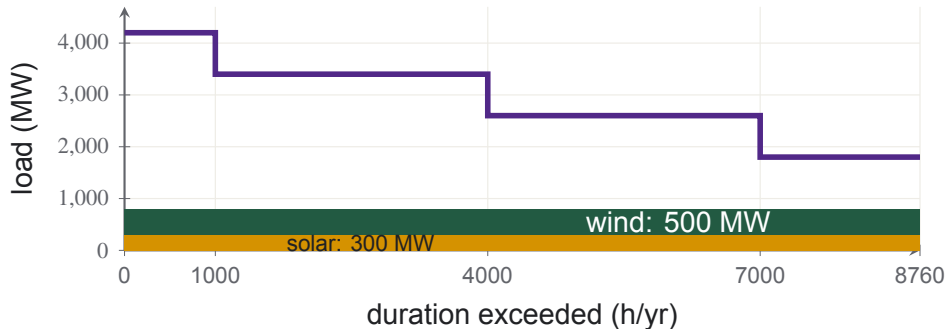
Wind: 500 MW
Solar PV: 300 MW

Candidate unit capacities

One CC unit: 543 MW
One GT unit: 211 MW

Place wind and solar

Begin with the two renewable capacity blocks given in the case



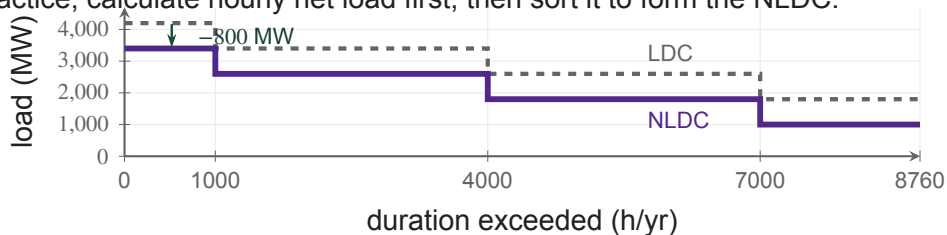
Assumption: the 800 MW renewable plan provides a constant output block across the duration curve.

Form the NLDC

Teaching case: subtract a constant 800 MW renewable-output block

$$\text{Net load} = \text{gross load} - 800 \text{ MW}$$

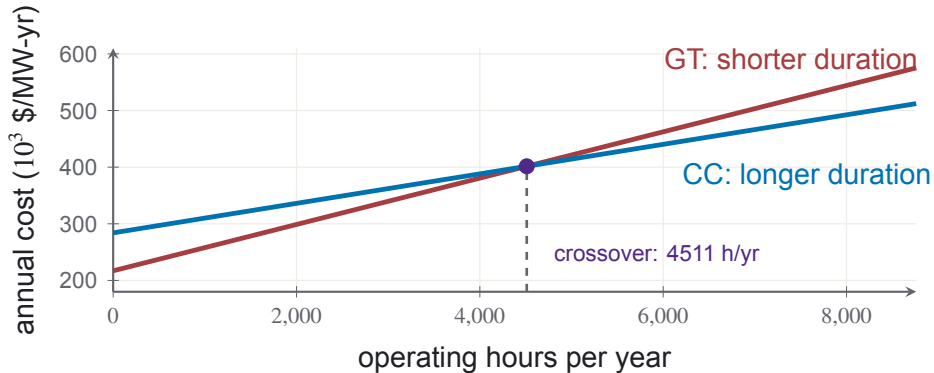
In practice, calculate hourly net load first, then sort it to form the NLDC.



The NLDC represents the demand served by CC and GT, with a 3400 MW peak.

Recall the GT–CC crossover

Part I already identified which dispatchable technology is lower by duration

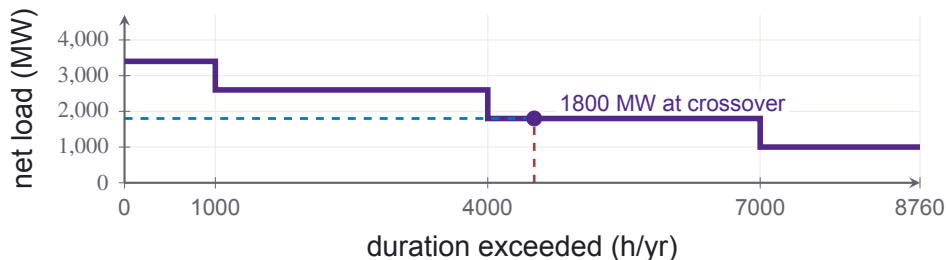


GT has the lower annual cost below 4511 h/yr. CC has the lower annual cost above 4511 h/yr.

Map crossover to MW

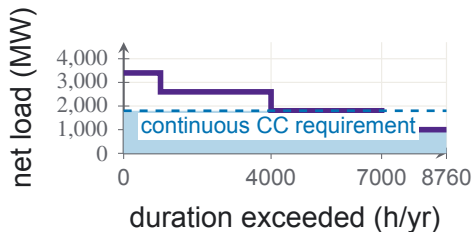
Read the net-load level at the crossover duration

At 4511 h/yr : net load = 1800 MW



The crossover duration becomes a capacity boundary on the NLDC.

Round CC units



CC calculation

Required capacity: 1800 MW

$$\text{Units} = \left\lceil \frac{1800}{543} \right\rceil = 4$$

Installed capacity: $4 \times 543 = 2172$ MW

Rounding adds 372 MW of CC capacity. This installed capacity is credited before GT sizing.

Round GT units for peak load

GT load-matching calculation

Net peak after renewables: $4200 - 800 = 3400$ MW

Required GT capacity: $3400 - 2172 = 1228$ MW

$$\text{Units} = \left\lceil \frac{1228}{211} \right\rceil = 6$$

Installed GT capacity: $6 \times 211 = 1266$ MW

installed: 3438 MW

CC: 2172

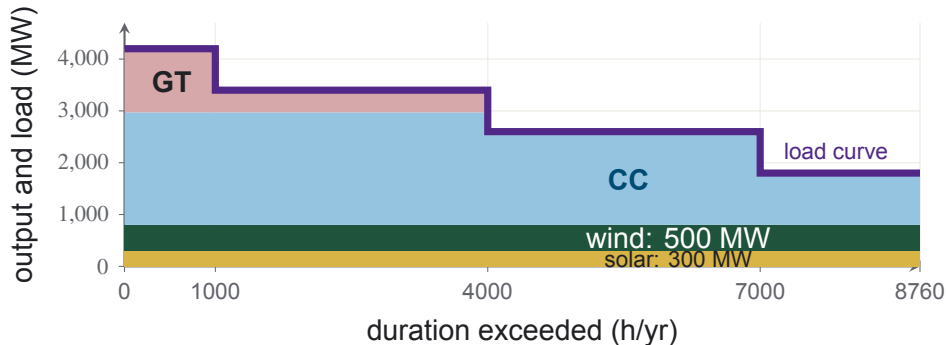
GT: 1266

required net peak: 3400 MW

Four CC units and six GT units provide 3438 MW for the 3400 MW net peak.

Check every load block

Installed capacity stays fixed while dispatched output follows the load level



The top of the generation stack matches the load curve in every duration block.

Add PRM after load matching

After load matching, compare credited capacity with the 15% target

1. Find the capacity shortfall

$$\begin{aligned}C_{\text{load}} &= 500 + 300 + 2172 + 1266 \\ &= 4238 \text{ MW},\end{aligned}$$

$$C_{\text{target}} = 1.15(4200) = 4830 \text{ MW},$$

$$\Delta C = 4830 - 4238 = 592 \text{ MW}.$$

2. Add whole GT units

$$N_{\text{add}} = \left\lceil \frac{592}{211} \right\rceil = 3,$$

$$N_{\text{GT,final}} = 6 + 3 = 9,$$

$$C_{\text{final}} = 4238 + 3(211) = 4871 \text{ MW},$$

$$\text{PRM}_{\text{final}} = 16.0\%.$$

Repeat the planning loop

The case illustrates a reusable screening sequence

1 Choose a duration boundary

Use the next crossover from the screening curves.

2 Map time to load

Read the net-load level at that duration on the NLDC.

3 Size load-serving units

Round CC and GT units to cover every load block.

4 Close PRM at the end

Compare credited capacity with the target. Add whole units to close any shortfall.

Complete load matching before the final PRM capacity check.

How the two lectures connect

L03 sets the capacity need, and L04 helps choose the technologies

Lecture 03

- **LDC:** load level and duration
- **PRM:** planning-capacity target

Lecture 04

- **Screening curves:** cost by operating hours
- **Crossovers + NLDC:** technology capacity blocks

duration + cost + adequacy $\longrightarrow$ candidate generation mix

Summary

Knowledge to carry forward

- **Annual-cost comparison:** fixed cost sets a screening curve's intercept. Variable operating cost sets its slope.
- **Technology selection:** crossover durations divide the operating-hour axis into ranges in which different dispatchable technologies have the lowest annual cost.
- **Net-load allocation:** mapping a crossover duration to the NLDC converts a cost comparison into technology capacity blocks.
- **Load-matching capacity:** round to whole units, then verify that every load block is covered.
- **Final adequacy check:** compare credited capacity with the PRM target. Add enough capacity to close any remaining shortfall.

Match the load first, then verify the mix against the PRM target and close the capacity gap.